

RAVINEESH SINGH

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🌐 LinkedIn • 🐙 GitHub • 🌐 Portfolio • </> LeetCode

Computer Science & Engineering undergraduate at IEST Shibpur with an **8.81/10 CGPA**, combining strong analytical problem solving with technical depth in **Data Structures & Algorithms, Java and C# development, Enterprise Data Pipelines**, and **Applied Machine Learning**. Experienced in designing scalable data ingestion workflows, training deep learning architectures (CNNs), and applying structured, secure engineering practices to deliver high-impact, customer-focused solutions for enterprise and financial services environments.

EDUCATION

Indian Institute of Engineering Science and Technology (IEST), Shibpur Bachelor of Technology in Computer Science and Engineering — CGPA: 8.81 / 10	West Bengal, India Aug 2024 – May 2028
Higher Secondary Education (Class XII) — UPMSP Uttar Pradesh Board of High School and Intermediate Education — Percentage: 83.6%	Uttar Pradesh, India Apr 2021 – May 2023
Secondary Education (Class X) — UPMSP Uttar Pradesh Board of High School and Intermediate Education — Percentage: 89.17%	Uttar Pradesh, India Apr 2019 – May 2021

FEATURED PROJECTS

Real Estate Price Predictor & Market Analytics Platform • ETL & Data Ingestion Pipeline: Engineered an automated web scraping and ETL pipeline in Python to extract, clean, and normalize 1,000+ multi-attribute property records from 99acres.com, ensuring schema integrity. • Feature Engineering & Machine Learning: Architected end-to-end data preprocessing workflows incorporating outlier detection, categorical encoding, feature scaling, and multi-algorithm regression (Linear, Ridge, Random Forest) to forecast valuations with high precision. • Business Intelligence & Market Insights: Developed exploratory data analysis (EDA) frameworks to model price elasticity, geographical valuation trends, and commercial demand indicators for data-backed decision making.	<i>Python, Pandas, Scikit-learn, SQL</i> 🐙 GitHub
AI-Driven Precision Agriculture: Weed Detection CNN • Dataset Engineering & Augmentation: Curated and annotated a domain-specific dataset of ~2,000 high-resolution agricultural images, applying image transformation and augmentation pipelines (zoom, shear, rotation) to prevent overfitting. • Convolutional Neural Network (CNN): Designed and trained a custom deep CNN architecture in Keras utilizing Batch Normalization and Dropout regularization (p=0.4) for high-accuracy feature extraction and real-time classification. • Model Optimization & Edge Deployment: Benchmarked model performance using Precision, Recall, F1-Score, and Confusion Matrix, targeting edge deployment to automate targeted herbicide spraying and drive sustainable agribusiness efficiencies.	<i>Python, Keras, TensorFlow, Deep Learning</i> 🐙 GitHub

TECHNICAL & PROFESSIONAL SKILLS

Programming Languages:	Python, Java, C#, C++, C, SQL, JavaScript (Basics)
Cloud & Modern Enterprise:	Enterprise Application Development (Java / C#), Cloud Concepts (AWS / Azure Fundamentals), RESTful APIs, ETL Pipelines, Relational Database Design (DBMS)
Data Science & AI / ML:	Convolutional Neural Networks (CNNs), Deep Learning, Predictive Modeling, Regression, Classification, Scikit-learn, Keras, TensorFlow
Data Analysis & Visualization:	Pandas, NumPy, Matplotlib, Seaborn, Business Intelligence (Power BI / Tableau concepts), Exploratory Data Analysis (EDA)
Software Engineering & Tools:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP with Java / C#), Git, GitHub, VS Code, Linux/Bash, Command Line / Terminal (CLI), Agile / Scrum, SDLC

CERTIFICATIONS & COURSEWORK

Command-Line & Terminal Fundamentals:	Completed a hands-on Terminal course covering shell navigation, file and directory operations, and core command-line utilities for efficient development workflows (Caddy).
Data Structures & Algorithms (DSA):	Comprehensive coursework in algorithmic problem solving, computational complexity optimization, dynamic programming, and tree/graph structures (Udemy).
Core Academic Coursework:	Data Structures & Algorithms, Design & Analysis of Algorithms, Object-Oriented Programming, Database Management Systems (DBMS), Operating Systems, Discrete Mathematics.

LEADERSHIP & COMMUNITY ENGAGEMENT

Core Member — Icche (Social Welfare Club, IEST Shibpur)	IEST Shibpur
Educational Outreach & Mentorship:	Volunteer 3 days weekly to teach foundational STEM and literacy to underprivileged K-12 students, fostering educational equity and measurable academic improvement.
Project & Event Management:	Lead cross-functional student volunteer teams to plan and execute campus-wide community welfare drives, logistics for annual Sports Days, and regional clothing/relief campaigns.